

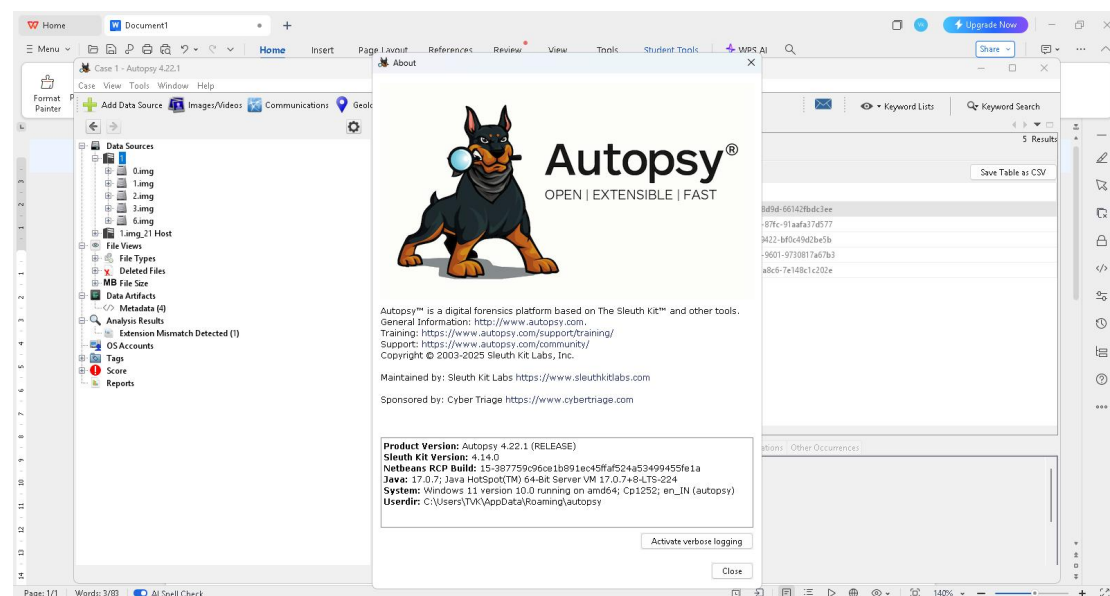
## Practical Sheet 1 Rhino Material

### 1.Question 1 (1 point)

To reward your efforts to get the GUI version of Autopsy running on your system, open Autopsy, select "Help" > "About" and copy the version info out of the framed box and paste it here.

If you are only able to run the web-UI version of Autopsy, paste the command-line output when starting Autopsy. You will get the point for a working GUI version and also for the web-UI. Your answer will be evaluated using regular expressions.

Ans: The version is Autopsy 4.22.1



### 2.

You are given 7 image files. To ensure the integrity of your investigation, please compute or extract the MD5 and SHA1 hash values for each of the image files. You get 0.25 points per correct hash value.

The answer needs to be given as a JSON array

Ans: [ {"filename": "0.img", "md5": "3225011e0c58c91a9161a53262e4d479", "sha1": "fcd5048778bc3e4a58c9c3f00c76b7ef83c0076e"},

{"filename": "1.img", "md5": "071a0cf9ca59284b25cade1615712571", "sha1": "6d06b3e60fd7093e0e2233345ac415351ec44971"},

{"filename": "2.img", "md5": "f804a8a3a172e425466e579267e32a90", "sha1": "cf64e674ae68bb71b1f4bbf48636e746c69c6d6b"},

{"filename": "3.img", "md5": "505cb5b050a1a98ee7b63d3ef9d6b67a", "sha1": "299c3f4ba5ccf4c341e4eaf04d292e18c8c5573"},

{"filename": "4.img", "md5": "72c218cc7b68bf3a07f3dc5d95a78d28", "sha1": "504eac0986be2226ffdeb267569c137bd7631a2b"},

{"filename": "5.img", "md5": "177beb6b68b68250368bcecab1e4d3e1", "sha1":

4aaf468346301baf6153b6c8f5bbc20746252cfd },

```
{ "filename": "6.img", "md5": "1170a8597547aa9771809fbb4d557ac0", "sha1":  
d4dc5aea39c501ee56cd6c75d72b8ed0df7de5bf" }
```

Description: To get the hash values of the images we need to upload the images in autopsy and then check the file metadata of each image file. There we can get the MD5 and SHA1 hash values for each image file.

The screenshot shows the Autopsy 4.22.1 interface. On the left, the 'Data Sources' pane shows a tree view with '1.img\_21 Host' selected. The main pane displays a 'Listing' table with columns: Name, Type, Size (Bytes), Sector Size (Bytes), Timezone, and Device ID. The table contains five rows of image files (0.img, 1.img, 2.img, 3.img, 6.img). Below the table, the 'File Metadata' pane shows details for '0.img', including Name, Type, Size, MD5, SHA1, SHA-256, Sector Size, Time Zone, Acquisition Details, Device ID, Internal ID, and Local Path.

| Name  | Type  | Size (Bytes) | Sector Size (Bytes) | Timezone      | Device ID                            |
|-------|-------|--------------|---------------------|---------------|--------------------------------------|
| 0.img | Image | 10240000     | 512                 | Europe/Berlin | f6f5ca60-55b7-4264-8d9d-66142fbd3ee  |
| 1.img | Image | 10240000     | 512                 | Europe/Berlin | b18a6599-7564-4b9c-87fc-91aafa37d577 |
| 2.img | Image | 10240000     | 512                 | Europe/Berlin | 590aff9e-3ddf-4182-9422-bf0c49d2be5b |
| 3.img | Image | 10240000     | 512                 | Europe/Berlin | 046c98a4-cdc3-46ad-9601-9730817a67b3 |
| 6.img | Image | 25600000     | 512                 | Europe/Berlin | cf6dfb0f-b6d1-4da6-a8c6-7e148c1c202e |

**File Metadata**

Name: /img\_0.img  
Type: Raw Single  
Size: 10240000  
MD5: 3225011e0c58c91a9161a53262e4d479  
SHA1: fcd5048778bc3e4a58c9c3f00c76b7ef83c0076e  
SHA-256: 31e1b13eee7f9992ebf8ef5b6067896f3117113456cc825a330bb5911f92dcf  
Sector Size: 512  
Time Zone: Europe/Berlin  
Acquisition Details: Unknown  
Device ID: f6f5ca60-55b7-4264-8d9d-66142fbd3ee  
Internal ID: 1  
Local Path: C:\Users\TVK\Downloads\Images\_for\_Practical\_Sheet\_1(Rhino\_Material)\images\images\images\0.img

But for 4.img and 5.img we can use Kali Terminal and we use md5sum and sha1sum commands to get the hash values.

```
(tvk@Kali)-[~/Downloads/images]  
$ md5sum 4.img > md5_hashes4.txt  
  
(tvk@Kali)-[~/Downloads/images]  
$ sha1sum 4.img > sha1_hashes4.txt  
  
(tvk@Kali)-[~/Downloads/images]  
$ md5sum 5.img > md5_hashes5.txt  
  
(tvk@Kali)-[~/Downloads/images]  
$ sha1sum 5.img > sha1_hashes5.txt  
  
(tvk@Kali)-[~/Downloads/images]  
$
```

3. You are given ".img" files. Which properties does this file format have?

Ans: It is a bit by bit copy of the original system/device.

4. Some of the image files are different from the others in one fundamental aspect influencing how you need to work with these images. Based on that difference, you can form two groups and assign every image file to one of these groups. Create a JSON array of arrays containing these groups. The order of elements inside the array is not relevant. This exercise is binary graded (correct/incorrect).

Ans:[  
["0.img", "1.img", "2.img", "3.img", "6.img"],  
["4.img", "5.img"]  
]

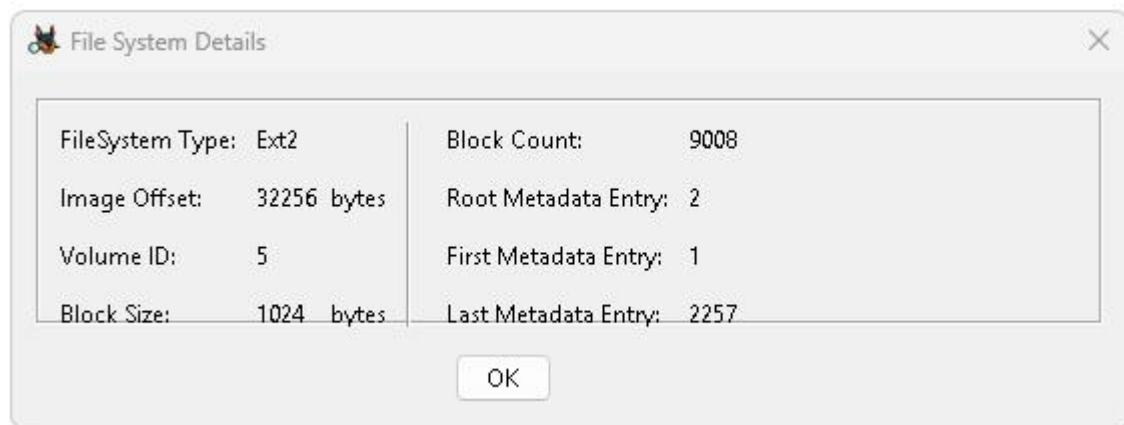
The 4.img and 5.img files are corrupt and have invalid partitions unlike other image files.

5. What are the differences here that require you to investigate those images differently?

Ans:One group of images contains valid Partitions,while others do not.

6. What is the file system format of all allocated partitions?

Ans.ex2

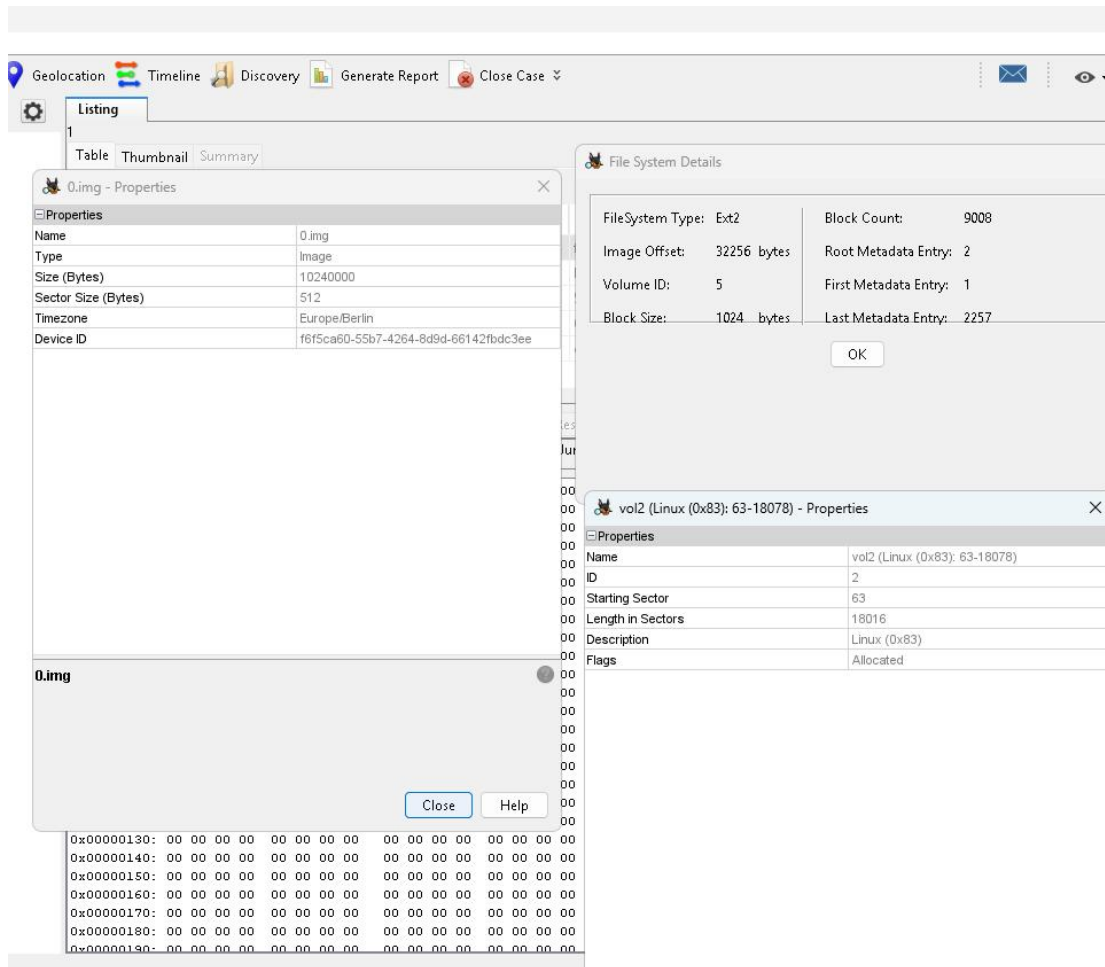


7. How are sector size, the number of sectors, the block size, and the number of blocks connected? Include these values of the partitions in your "proof of work" file.

Ans:Sector Size:512  
Number of Sectors:18016  
Block Size:1024  
No of Blocks:9008

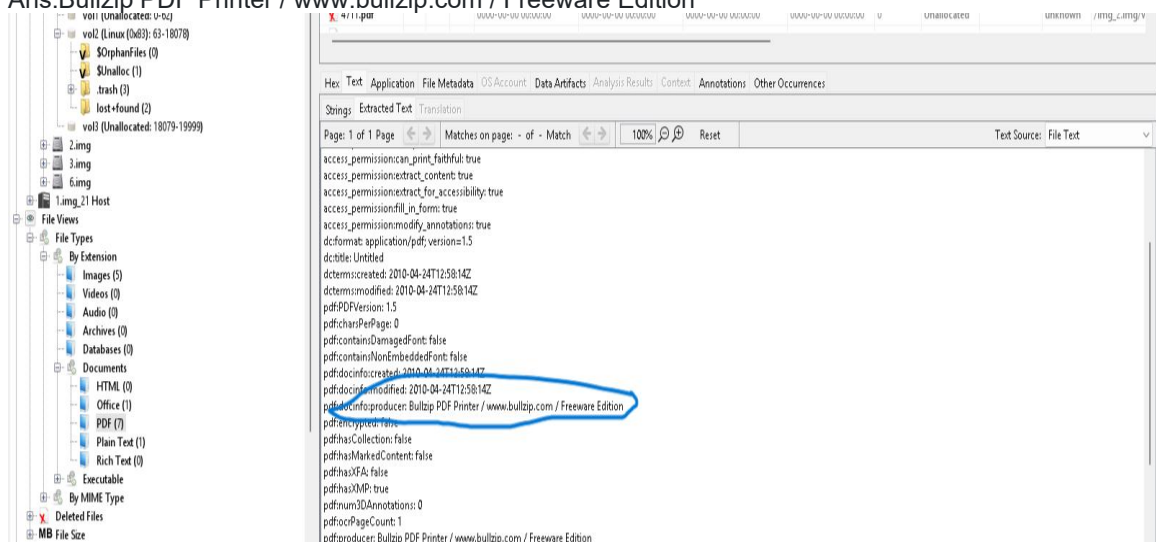
$18016 * 512 = 9224192$   
 $9008 * 1024 = 9224192$

Hence the product of the first and second equals the product of third and fourth.



8. Which program was used to produce the allocated PDF file? Your answer will be evaluated using a regular expression.

Ans: Bullzip PDF Printer / www.bullzip.com / Freeware Edition



In the Text section of the image file we can find the program which was used to create the pdf file.

9. Give a JSON-array of all files that you found containing rhinographic material with their filename and both an **MD5**-hash and a **SHA1**-hash. One point will be deducted per missing file, but you cannot get less than 0 points. If you give incorrect hash values, the file will be ignored.

The answer needs to be given as a JSON array.

Ans: We need to search files containing rhinographic material so manually opening each and every file is one of the approaches. For the Hash Values MD5 hash value can be obtained by inspecting the properties of the files.

The files which I got are:-

```
[
{"filename": "image0.png", "md5": "3e85149afe34a977a18b33b04caa0fce", "sha1": "326c70a80bd66c59ab3565dd7f0786cd583af9c0"},

{"filename": "image0.png", "md5": "d499f2211d7047b443e43a87a4b85c16", "sha1": "631053f9c0dc97b73a84b323f089dcad4d586b20"},

{"filename": "secret.doc", "md5": "b3c474fa9cfb5c749aa67d4e20b97014", "sha1": "c59e918356a9b6ef579cae5b15ecec67445aebf7"},

{"filename": "74656.pdf", "md5": "dc81af3e404446a02c8b76d5a8a710bd", "sha1": "3aadd4d829d27d22379d74961ab8c0ced89e19ed"},

{"filename": "f0003706.pdf", "md5": "21de5bf08599392976f574392400eeac", "sha1": "77f43caf1e1ff9e7618a63cf41f8681cf784919a"},

{"filename": "f0015503.txt", "md5": "19dae2b2c6eac4c9c845d66d6d960df", "sha1": "3c08a474bf53330753503d60cf4e2eef55b17e26"}
]
```

The screenshot shows a file analysis tool interface. On the left, a tree view displays 'Data Sources' with a folder '1' containing files '0.img', '1.img', '2.img', '3.img', and '6.img'. Below this, 'File Views' are categorized by extension (Images: 5, Videos: 0, Audio: 0, Archives: 0, Databases: 0, Documents, Executable) and by MIME Type (application, image, text). 'Deleted Files' are listed as MB 50 - 200MB (0), MB 200MB - 1GB (0), and MB 1GB+ (0). 'Data Artifacts' include Metadata (4), Analysis Results (Extension Mismatch Detected (1)), OS Accounts, Tags, Score, and Reports.

The main pane shows a table of files with columns: Name, S, C, O, Modified Time, Change Time, Access Time, Created Time, Size, Flags(Dir), Flags(Meta), Known, and Location. The table contains 5 results:

| Name       | S | C | O | Modified Time           | Change Time             | Access Time             | Created Time        | Size  | Flags(Dir) | Flags(Meta) | Known   | Location          |
|------------|---|---|---|-------------------------|-------------------------|-------------------------|---------------------|-------|------------|-------------|---------|-------------------|
| image0.png |   | 1 |   | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00 | 39497 | Allocated  | Allocated   | unknown | /img_1.img/vol_vc |
| image0.png |   | 1 |   | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00 | 39497 | Allocated  | Allocated   | unknown | /img_1.img/vol_vc |
| image0.png |   | 0 |   | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00 | 25497 | Allocated  | Allocated   | unknown | /img_2.img/vol_vc |
| nimda.jpg  |   | 0 |   | 2009-02-14 00:31:32 CET | 2009-02-14 00:31:32 CET | 2009-02-14 00:31:31 CET | 0000-00-00 00:00:00 | 1     | Allocated  | Allocated   | unknown | /img_3.img/vol_vc |

Below the table, a preview of a file is shown. The file is a black and white image of a rhinoceros, which is the 'rhinographic material' mentioned in the question. The preview includes a zoom level of 41% and a 'Reset' button.

Autopsy 4.22.1 interface showing the Listing view for an Office file named secret.doc. The table displays file metadata including Name, S, C, O, Modified Time, Change Time, Access Time, Created Time, Size, Flags(Dir), Flags(Meta), Known, and Location. The file is located at /img\_0.

| Name       | S | C | O | Modified Time           | Change Time             | Access Time             | Created Time        | Size  | Flags(Dir) | Flags(Meta) | Known   | Location |
|------------|---|---|---|-------------------------|-------------------------|-------------------------|---------------------|-------|------------|-------------|---------|----------|
| secret.doc |   |   | 0 | 2009-02-14 00:31:32 CET | 2009-02-14 00:31:32 CET | 2009-02-14 00:31:32 CET | 0000-00-00 00:00:00 | 30212 | Allocated  | Allocated   | unknown | /img_0   |

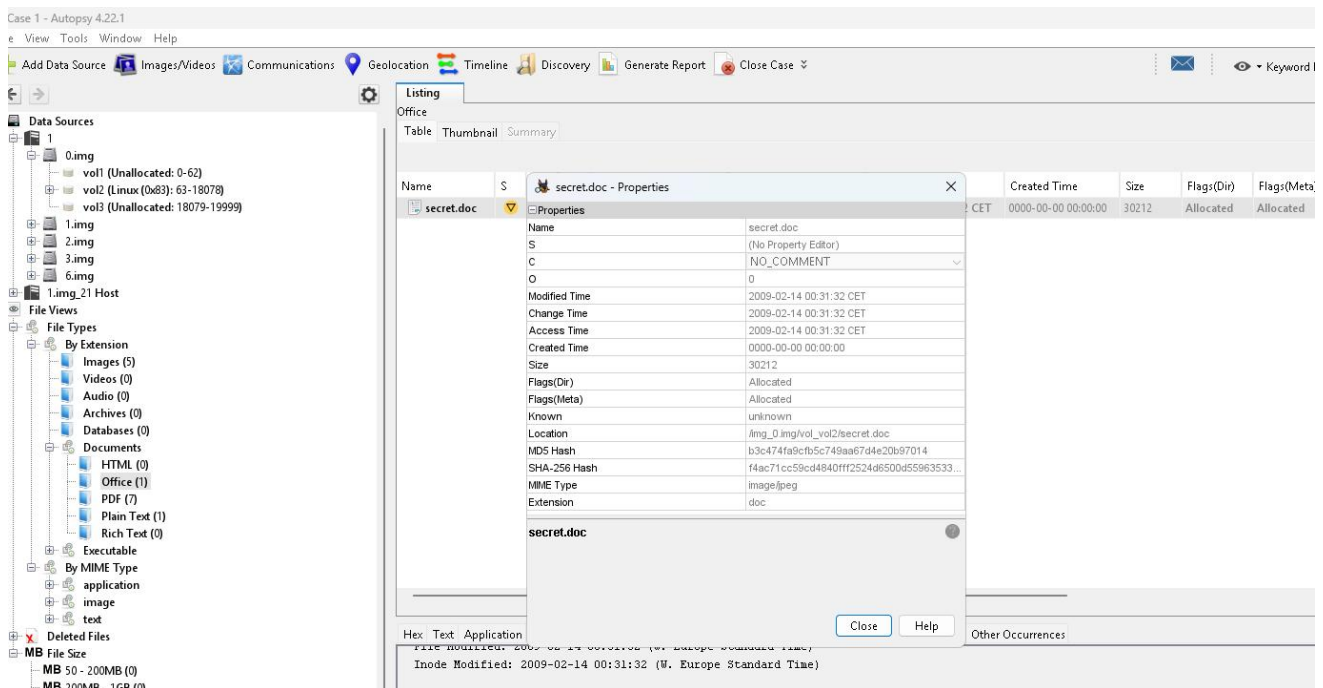
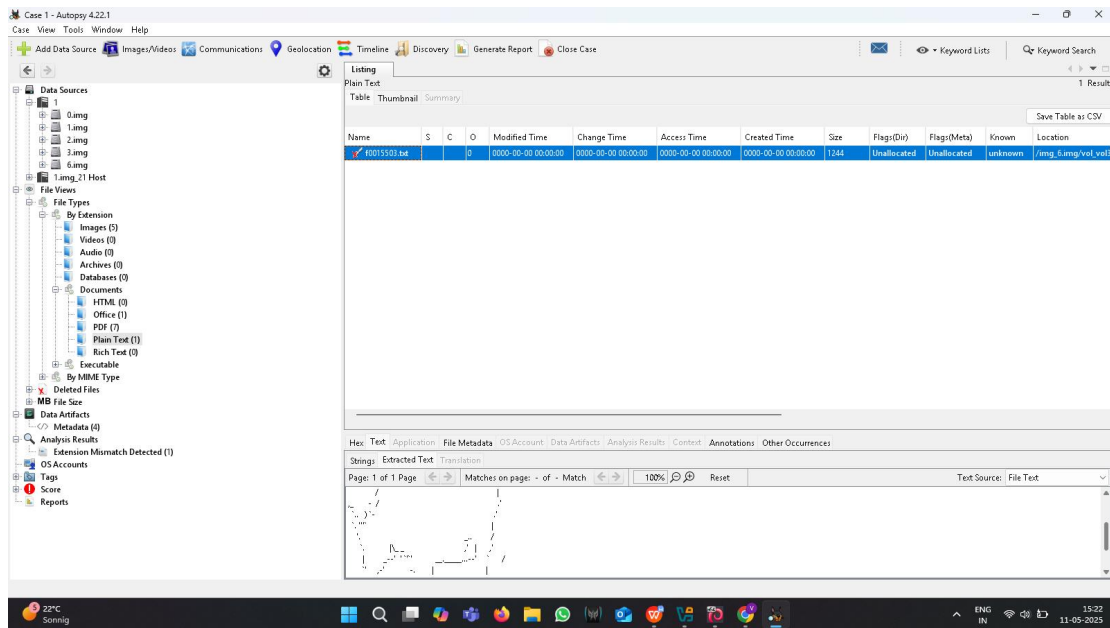
The bottom pane shows a hex view of the file content, which is mostly blacked out for redaction.

Autopsy 4.22.1 interface showing the Listing view for a PDF file named 74656.pdf. The table displays file metadata including Name, S, C, O, Modified Time, Change Time, Access Time, Created Time, Size, Flags(Dir), Flags(Meta), and Known. The file is located at /img\_0.

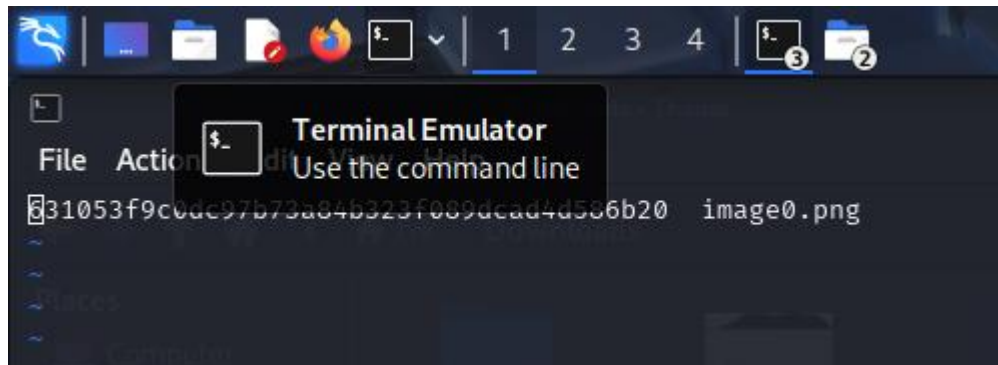
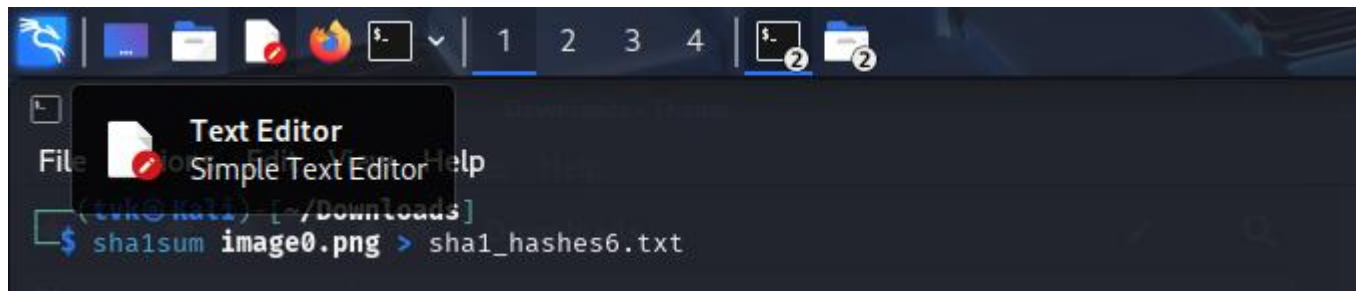
| Name         | S | C | O | Modified Time           | Change Time             | Access Time             | Created Time        | Size  | Flags(Dir)  | Flags(Meta) | Known   |
|--------------|---|---|---|-------------------------|-------------------------|-------------------------|---------------------|-------|-------------|-------------|---------|
| 74656.pdf    |   |   |   | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00 | 0     | Unallocated | Unallocated | unknown |
| 74656.pdf    |   |   | 1 | 2009-02-14 00:31:32 CET | 2009-02-14 00:31:33 CET | 2009-02-14 00:31:32 CET | 0000-00-00 00:00:00 | 49948 | Allocated   | Allocated   | unknown |
| 74656.pdf    |   |   |   | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00 | 0     | Unallocated | Unallocated | unknown |
| 74656.pdf    |   |   | 1 | 2009-02-14 00:31:32 CET | 2009-02-14 00:31:33 CET | 2009-02-14 00:31:32 CET | 0000-00-00 00:00:00 | 49948 | Allocated   | Allocated   | unknown |
| 4711.pdf     |   |   |   | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00 | 0     | Unallocated | Unallocated | unknown |
| 4711.pdf     |   |   |   | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00 | 0     | Unallocated | Unallocated | unknown |
| f0003706.pdf |   |   | 0 | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00 | 50972 | Unallocated | Unallocated | unknown |

The bottom pane shows a hex view of the file content, which is mostly blacked out for redaction.





For SHA 1 hash values we can follow the same method from question 2 where we use kali Terminal to get the hash values. So first we need to export the individual docs and png files and use these files in kali Terminal to get the SHA-1 hash values.



Similarly we can follow this for the remaining png and doc files and extract SHA-1 hash values.

10. In what timespan are the valid (not just zeroes) timestamps of the files that you carved? You get one point for each correct date. Use the format "DD.MM.YYYY - DD.MM.YYYY".

Ans: After inspecting the img files I got 2 different timestamps for 2 files

14-02-2009 - 14-02-2009

24-04-2010 - 24-02-2010

The second timestamp was only for 1 particular pdf file remaining all other files had the first timestamp majorly.



Case 1 - Autopsy 4.22.1

Case View Tools Window Help

Geolocation Timeline Discovery Generate Report Close Case

Listing

Table Thumbnail Summary

| Name       | S | C | O | Modified Time           | Change Time             | Access Time             | Created Time        | Size | Flags(Dir) | Flags(Meta) | Known   |
|------------|---|---|---|-------------------------|-------------------------|-------------------------|---------------------|------|------------|-------------|---------|
| secret.doc |   |   |   | 2009-02-14 00:31:32 CET | 2009-02-14 00:31:32 CET | 2009-02-14 00:31:32 CET | 0000-00-00 00:00:00 | 2012 | Allocated  | Allocated   | unknown |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings: Extracted Text Translation

Page: 1 of 1 Page Matches on page: - of - Match 100% Reset Text Source: File Text

JFF  
z00000005 204629  
\$for  
%&("456789CDEFGHJSTUVWXYZcdefghijstuvwyz.  
#JR

Case 1 - Autopsy 4.22.1

Case View Tools Window Help

Geolocation Timeline Discovery Generate Report Close Case

Listing

Table Thumbnail Summary

| Name         | S | C | O | Modified Time           | Change Time             | Access Time             | Created Time        | Size  | Flags(Dir)  | Flags(Meta) | Known   | Location       |
|--------------|---|---|---|-------------------------|-------------------------|-------------------------|---------------------|-------|-------------|-------------|---------|----------------|
| 74656.pdf    |   |   |   | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00 | 0     | Unallocated |             | unknown | /img_1.img/vol |
| 74656.pdf    |   |   | 1 | 2009-02-14 00:31:32 CET | 2009-02-14 00:31:32 CET | 2009-02-14 00:31:32 CET | 0000-00-00 00:00:00 | 49948 | Allocated   | Allocated   | unknown | /img_1.img/vol |
| 74656.pdf    |   |   |   | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00 | 0     | Unallocated |             | unknown | /img_1.img/vol |
| 74656.pdf    |   |   | 1 | 2009-02-14 00:31:32 CET | 2009-02-14 00:31:32 CET | 2009-02-14 00:31:32 CET | 0000-00-00 00:00:00 | 49948 | Allocated   | Allocated   | unknown | /img_1.img/vol |
| 4711.pdf     |   |   |   | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00 | 0     | Unallocated |             | unknown | /img_2.img/vol |
| 4711.pdf     |   |   |   | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00 | 0     | Unallocated |             | unknown | /img_2.img/vol |
| f0003706.pdf |   |   | 0 | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00     | 0000-00-00 00:00:00 | 50972 | Unallocated | Unallocated | unknown | /img_2.img/vol |

Save Table as CSV

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings: Extracted Text Translation

Page: 1 of 1 Page Matches on page: - of - Match 100% Reset Text Source: File Text

access.permissions.permissions: true  
dcformat: application/pdf; version=1.5  
dc:title: Untitled  
dcterms:created: 2010-04-24T12:58:14Z  
dcterms:modified: 2010-04-24T12:58:14Z  
pdf:PDFVersion: 1.5  
pdf:char:PerPage: 0

11. Autopsy will likely assign one file the score "suspicious".

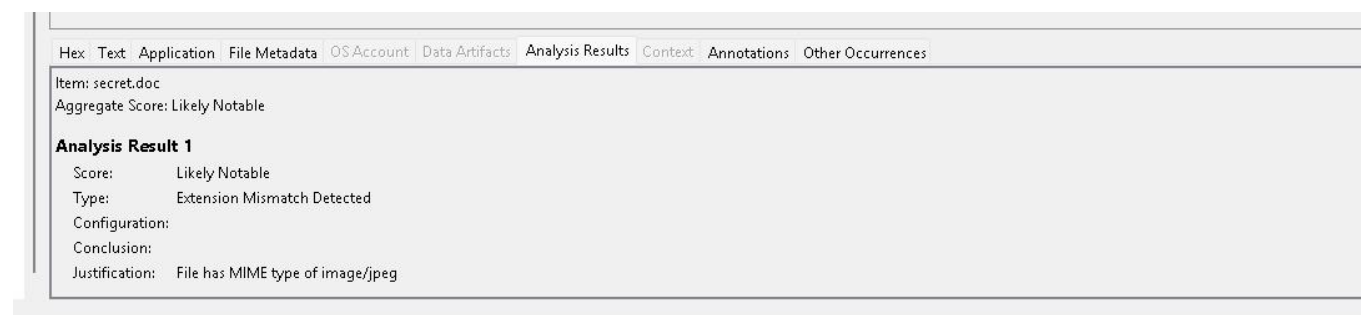
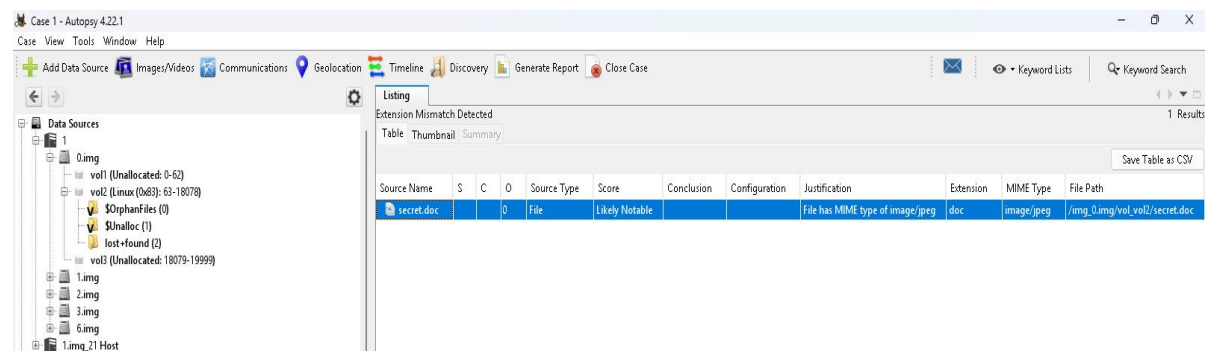
a) What is the filename of the suspicious file? (0.5 points)

Ans: secret.doc

b) How did Autopsy come to this conclusion? (1.5 points)

(It is crucial to include relevant char sequences and words here such that a regex can match your answer.)

Ans: In the Analysis Result it showed that Extension Mismatch Detected and it justifies by stating File has MIME type of image/jpeg.



12. One image seems corrupted. What is here most likely the case?

Ans: The forensic expert doing the image used a broken software write blocking tool that corrupted some bits.

13. Which groups of users were allowed to modify the files on the original system?

Ans: The File owner